Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Name: _____

Student ID: _____

Quiz 7: 09/29 Content

MATH 1271

10/01/2026

This quiz is worth 5 points and consists of 1 questions. You have 5 minutes to complete the quiz.

Question	Points
1	5
Total:	5

1. (5 points) The graph of a function $f(x)$ is given below:

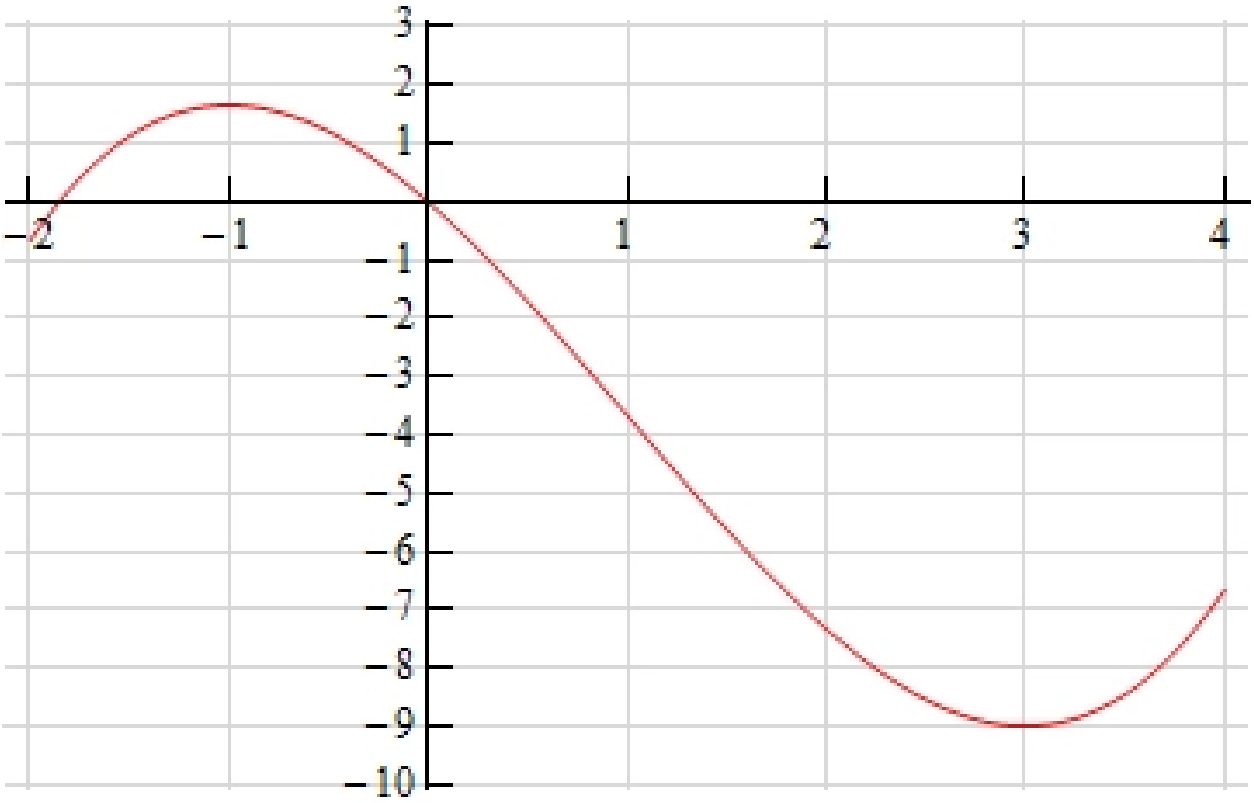


Figure 1: Graph of $f(x)$

Use this to sketch the graph of the derivative $f'(x)$ on the same axes.

Solution: We note that $f(x)$ is:

1. Increasing on $(-\infty, -1)$
2. Stationary at $x = 1$
3. Decreasing on $(-1, 3)$
4. Has an inflection point somewhere around $x = 1.5$
5. Has a stationary point at $x = 3$
6. Increases on $(3, \infty)$

which implies that

1. $f'(x)$ must be positive on $(-\infty, -1)$
2. $f'(x) = 0$ when $x = 1$
3. $f'(x)$ must be negative on $(-1, 3)$
4. $f'(x)$ must have a stationary point somewhere around $x = 1.5$
5. $f'(x)$ must be zero at $x = 3$
6. $f'(x)$ must be positive on $(3, \infty)$

which we can combine together to get

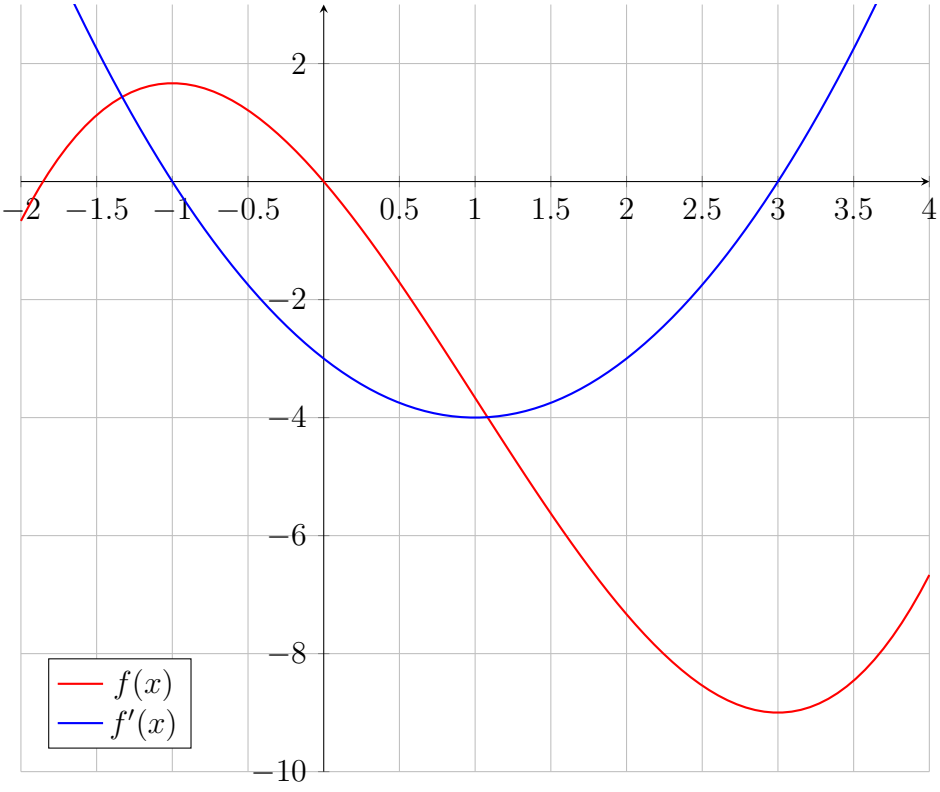


Figure 2: Graph of $f(x)$ and $f'(x)$